Io, gli Elefanti e la Teoria dei Giochi

Formal Models in Comparative Politics and International Relations

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Abstract

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Introduction

In this paper I wish to describe an intuition regarding the way game thoery has yet formalized some interactions between agents. Game thoerists consider surprises and shocks as the rationale for an irrational behaviour. I believe that the following paragraphs will convice the audience about the value of this contribution in two sense; first, I hope the description provided will be further used and applied. Second, the intuition about the role of emotions adds a new perspective to the literature about bounded rationality.

The intuition regards the field of animal behaivour, that has drown a lot from game thoery. Even though it may appear far from what economists study, still, it deals with the problems that game thoerists often face: the first is to represent the processes that happen in agents mind, this means taking into consideration and "measuring" variables that we do not observe. The second problem deals with the creation of a formal idea about this processes. The last problem connects this idea with the observed behaviour, and estabilishes an equilibrioum, thus a state of the world with particular properties [1].

In order to ease the description and to make the models closer to a possible application, I expolit a singular situation described in Safina, 2015. This book is an amazing challenge for every person who has ever had to face a game thoery excercise. To support his thesis, the author provides the reader detailed descriptions of animal interactions. All species selected belong to the realm of "cleaver" animals, those animals capable of emotions and complex cognitive processes. I believe that each of these interactions is a good game theory excercise, since they forces the game thoerist to apply its knowledge out of his area of expertise. This book also provides a simplified setting for these exercises, since animals constitute the perfect agents in game thoery. They are complex enough for beign represented as rational players, but they are not as complex as humans, so that most of the assumptions and simplifications, have more grasp into reality. Under this respect, elephants appear to be the perfect application. There is plenty of evidence about their behaviours, and there is plenty of evidence about the fact they act in accordance with a *mind*. According to Safina they are almost as humans, and they often interact with humans.

I will formalise my intuition thorugh a Dyaciclic Graph(DAG). This tool is rarely used in Game thoery, however is more often used in empirical studies. Since I wish to norrow down the gap between thoery and practice, I believe that it perfectly suits my research design. Additionally, provide a more transparent version of the analysis, clarifiyng the variables of interests and the limits of the assumptions.

The final aim of this article is to clarify a contradiction between two similar situations that game theory formalises in different ways. The models presented come from the signalling literature.

Beyond Words

Carl Safina is a biologist and wrote a world-famous essay (Safina, 2015) about animals' minds. He argues that until now, biologists have refused to deepen the study of animals' minds and souls and that is high time they did that. The curious thing about this book is that, in doing so, he tries to bring to readers' attention a problem quite close to game theory. He tries to rationalise the behaviour of some animals, arguing that their behaviours may be explained if we accept to consider them "intelligent", thus guided by a mind and

[1] questa parte va riscritta. voglio far vedere che è inutile cercare la spiegazione all'esterno perché potrebbe benissimo essere all'interno della mente

sensitive to emotions. This book is a tentative to take animals mind at the same level of human minds.

The Problem

This section aims at claryfing the setting to which the game thoery models refere, as well as making the reader aware of the basic intuition. In the examples Safina describes in his book (Safina, 2015) animals seem to be driven by rationality and emotions, as much as humans are. These are brief summaries of the two episodes that really happend. Both interactions between humans and elephants happened in a wildlife park in Africa. In both, the surrounding environment appears to be similar,

Case 1:

A staff member of a wildlife park in Africa reports an attack by an elephant on a tourist. The scene described clearly highlights the signals that the elephant sends to the tourist. It also clearly identifies the reasons why it is logical to believe that the elephants intentionally avoided hurting the tourist. In particular the staff member reports signs on the ground that undoubtedly mark the intention to halt the charge zagainst the tourist moments before impact. We can formalise this game though a signalling game in which the elephant sends a signal to threaten the tourist but doesnt have any intention of hurting him.

Another situation, recorded in the same environment, shows how similar conditions can produce an opposite pattern of behaviour.

Case 2:

In the same wildlife park in Africa, a rescue group was involved in a mission to reach a shepherd who was badly hurt by an elephant. When the staff approached him, they reported that the shepherd stopped them from harming the animal. He argued the elephant had kept him safe during the night. From the behaviour of the animal it appeared evident what was going on in his mind. After the attack, the elephant soon realised it had misinterpreted the signals from the shepherd, and after hurting him, the animal took care of him until the rescue group reached the place of the accident. Here, the same signalling structure of the first example breaks down: the equilibrium is reached but the elephant attacks. However, the event suggests a richer model of rationality, where the animal is able to recognise an "error".

Taken together, these cases expose a conceptual gap between real-world behaviour and the assumptions of rational-agent theory. OR Both scenarios can be framed within game theory, yet each reveals elements of cognition and emotion that standard models overlook.

Parallel with Game Theory

I believe game theory analysis accurately represents most of the aspects of both cases. For example, the fact that the animal starts a conflict with humans and tend to be pacific with most of other creatures is due to the fact that humans are one of the few threats they face in nature. Game theory correctly models agents as driven by incentives, it does not take into account interactions in which the agent acts violently because *it has a bad nature*. On the same line, game theory models credible threats as equilibria depending on the information available to players: not the turist nor the sheperd intended to porpusly harm an animal that may easly kill them, for example. Additionally, there are plenty of models that suite each of the two situations. For instance, as we will see, some models explain the behaviuor of the first case thorugh the concept of mimic: the elphant knows he is not going to do any harm to the tourist. However, the turist cannot distinguish between an animal with good or bad intentions, and the elephant exploits this weakeness for defense porposes. Thus both animals and humans may be depicted as rational agents, since their behaviour appears to be consistent with the environment and with a broader goal, if not with an aim in mind. Both appear to be able to communicate with each other, indicating that both species are adapting to signals from others and that they can act as informed agents. And both seem to be aware of each other's strengths, at least partially (Schelling, 1960; Smith, 1973). Again, in both cases, the animal seems "pacific" and willing to avoid or prevent conflict, at least at a certain point in time.

However, I believe we are still missing something. To describe these situations more accurately, we should introduce a new feature. Why do we see two different behaviours then? I believe these parallels also reveal what is absent from our theoretical framework. A possible explanation is that the two human beings differed from each other in some feature that was relevant to the context. On the other side, the same explanation may well be applied to the animals, which may be considered to have their own "personalities" (as Safina well explains in his book).

Representative Agents

In both cases analogies between men and animals do not appear to be crystal clear. However, most of the aspects they differ are never represented in game theory formalisations. Additionally, game theory is often used to study nations behaviours, this means that only the crucial features of agents need to be represented.

Thus, under the descriptive limits of these games and under their assumptions, I believe no harm is caused when I use the terms human, elephant and agent as synonyms. I will differentiate among these agents only in those cases in which it necessary, and with due advertisement to the reader. I will therefore continue with these examples in the belief that they will shed light on the way we theorise a broader set of agents and interactions. These examples serve both to reinforce the empirical evidence presented by Safina and to contribute to the theoretical assumptions underlying game-theoretic models or to provide a new equilibrioum concept.

Literature Review

Even though game theory has been applied to natural sciencie for a long time (Smith, 1973), interestingly, there is no poaper outthere that use animals as simplified examples of agents. I guess the reason is that the field are far from each others, and researchers of the two fields share even less common things. However, a reason why game theory has never been applied to animal minds yet, might be that since its origins, the field of biology has been slightly reluctant in considering the commonalities between human and animals minds.

Causal Inference

This reasonings naturally invites comparison with frameworks that model how hidden factors influence observed outcomes. Causal inference (Pearl, 2009) suggests that in order to answer this question we may also think of a "variable" that we are mistakenly excluding from the framework. My guess is that in this context this missing variable exists.

Application of Potential Outcome Model

This section is intended to formalise the intuition thorugh a DAG. Before diving deep into the DAG, I will briefly explain how does the concept of potential outcome applys to this situation, so that non expert readers will comfomfortably go through the next paragaphs.

A potential outcome is a counterfactual. Thus, it is a scenario that we do not observe; we think of it as being the closer and most realistic scenario that might have happened in case the actual one didn't.

In the case of the two intaractions I presented, each of them may be considered the counterfactual scenario of the other. Since counterfactuals do not exist, we use real and observed examples to mimic them. The underlieng belief, is that no research is as good as reality in producing examples of what might have happened. If we trust reality and observation of reality, the role of the researcher is limited to selecting the most appropriate counterfactual from those offered by natural world.

Hypothesis: under no effect of emotions, the case offered by the elephant and the sheperd would have been identical to the one offerd by the other elephant and the tourist.

DAGs

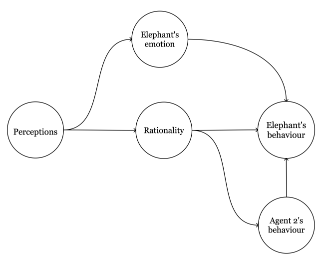

Figura 1: Un'immagine JPG inclusa in LaTeX.

Model 1 - Surprise as Shock

Citazione Battigalli su perfect and imperfect information

The decisive variable in this situation may be the element of surprise (Schelling, 1960). The two versions of the elephant-human encounter, the one involving the shepherd and the one involving the tourist, are analogous. In the tourist scenario, there is no surprise: the elephant is aware that humans might approach and disturb it, and it knows where to find them.

The situation involving the shepherd, however, contains the element of surprise, which may justify a different behavioral response, namely, the attack. What surprise alone does not explain, however, is why the elephant protected the shepherd after the attack.

To justify this behaviour we introduce empathy, force that shapes the outcome of the game.

However, before jumping to the model, we must ourselves what does empathy have to do with the first case. We may argue that empathy weakens elephants signal. Yet empathy also functions as a force that regulates the intensity of the signal itself.

Is it therefore reasonable to suggest that, in the second case, empathy may also be the mechanism that modulates the variability of the signal? What is the relationship between these two forces?

Now that the cause has been established, we can turn to the effects. We observe two primary effects: that of surprise and that of empathy.

Model 2 - Vote Buying

Paper di Grosier e Snyder 1996

In this section I propose a readaptation of a famous model (Groseclose and Snyder, 1996). I intend to expoilt the genious equilibrioum authors found to support the intuition coming from the DAG with another nerrative. In doing so, I believe we are improving the previous model and we are getting closer to the final aim of representing a real world event.

In their article (Groseclose and Snyder, 1996) the authors explain why in the US Congress the number of votes often exceedes the minimum required to make a bill pass. Thay argued that this is due to the interaction between lobbying groups competing for support of different bills. These groups *bribe* Congressmen to convince them.

The basic intuition from the paper comes from this example:

"The legislature contains seven members, all indifferent between the two proposals, that is, $v(i) = 0$ for all i. Since B will move second and attack the weakest part of As coalition, A will offer the same bribe, a, to all legislators he bribes. Let $m + 4 \geq 0$ be the size of the coalition A bribes. If $m = 0$, then B can defeat x by paying $a+\varepsilon$ to one member of As coalition. Thus, A must set $a \geq W_B$ to win, and As total bribes are then $4W_B$. If $m = 1$, then B must pay $a + \varepsilon$ to two members of As coalition to defeat x. Thus, A must set $a \geq W_B/2$ to win, and his total bribes are then $5W_B/2$. This is less than $4W_B$, so A is better off buying a supermajority of size 5 than a minimal winning coalition. If $m = 2$, then B must pay $a + \varepsilon$ to three members of As coalition, so A must set $a \geq W_B/3$ to win, and his total bribes are then $2W_B$. In fact, A minimizes his total payments by bribing all seven legislators. Here, $m = 3$, and As total bribes are $7W_B/4$. Clearly, there is an equilibrium where x wins if and only if $W_A \geq 7W_B/4$. If $W_A < 7W_B/4$, then there are

only equilibria in which no bribes are paid and s wins." (Groseclose and Snyder, 1996)

Representation through Vote Buying

Consider a council composed of several members. These members are not strategic players in the game but rather represent passive agents whose votes can be influenced. There are two active players, conceptualized as two groups, each seeking to "buy" the votes of the council members in order to determine the policy agenda. Each group possesses a budget, and bills are voted on by the council according to a predefined rule, assumed here to be the simple majority rule. This setup corresponds to the classical *vote-buying game* as formalized by (Groseclose and Snyder, 1996).

The objective of this analysis is to adapt this framework to the so-called *elephant-shaped* case. This adaptation allows the equilibrium structure of the original model to be employed within a cognitive and emotional context.

- Let K denote the total number of *thoughts*.
- The two players represent *opposing emotions* that regulate the internal dynamics of the elephants mind. "Setting the agenda" refers to determining the behaviour.
- Each players *budget* corresponds to the *intensity* of its emotion. A higher emotional intensity enables the emotion to "purchase" a larger number of thoughts.

Notation:

Players: $N = \{X, Y\}$ or $N = \{Emotion1, Emotion2\}$.

Histories: $H = \{\varnothing, \{x\}, \{x, y\}\}$. Where x and y indicate vectors of payments in the standard setting. Emotions use a unit of payment (emotional intensity) to corrupt a thought (or a judgement).

Player function: $p(\varnothing) = X; \quad p(x) = Y$

Actions: $A(\varnothing) = \{x\}; \quad A(x) = \{y\}\ \forall\, x$.

Each player (emotion) receives V_i if elephant acts accordingly.

Each player loses $\sum_{j=1}^{k} x_j$ for each unit of intensity spent.

This adapted framework enables a formal representation of conflict as a strategic interaction between emotions competing for cognitive control. Each emotion allocates its limited intensity across thoughts, attempting to secure a majority sufficient to determine the elephants mental and behavioral trajectory. The resulting equilibrium can be interpreted as a stable emotional configuration in which neither emotion can improve its outcome given the others allocation strategy. Let μ denote the minimal number of thoughts required to *set the agenda*, that is, to achieve behavioral dominance of one emotion over the other. Then:

$$\mu = \frac{k+1}{2}$$

Structure of the Paper

1. Introduction

   (a) I will copy and paste most of the parts of this research proposal. It seems to me like a good introduction.

   (b) Brief intro to clarify some concepts to readers coming from economics: Safina's book, Safina's work/life, what are elephants and why do they matter

   (c) The two examples (case 1 and case 2) from the book, (*with proper citation*)

   (d) I will present the problem (*thus the contradiction I believe I discovered*), along with a longer explanation of it. I will also draw a more precise connection with game theory, *maybe I will use bullet points*

2. Causal Inference
   I will expand on the inference problem from more perspectives. I will clarify that the cognition of elephants is just an hypothesis, and I will state clearly this hypothesis. I will draw the DAG for the problem coming from biology. I will draw the DAG for the game theory problem I wish to describe, along with its assumptions.

3. Models

   (a) Model nř1: perfect information and complete information for the case 1 from Safina's book

   (b) Model nř2: perfect information and complete information for the case 2 from Safina's book

   (c) Model nř3: According to what I come up with in the first part, I will try to put together a model that comprehends both of the two examples.

4. Models on EGT

   (a) Introduction to EGT (*history plus main concepts*)

   (b) Model nř4: Empathy is an ESS. With a good amount of luck I will by able to draw a model (*maybe*)

5. Conclusion:
   I will make sure that from this part will not miss a general summary, a summary of the part coming from biology, a summary of the conclusions and of the resuls.

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